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Operating Systems
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Lab 3 Report

Background:

The purpose of this assignment was to evaluate the behavior of three page replacement algorithms FIFO, LRU, and OPT using a fixed reference string of 10,000 page accesses and four different physical memory sizes: 128, 256, 512, and 1024 frames. By calculating the page-fault rate at intervals of 2000 references, we can observe how each algorithm responds to increasing memory availability and how closely they approximate the theoretical optimal behavior.

Analysis:

When the system is limited to 128 frames, both FIFO and LRU perform almost identically, each producing extremely high total fault counts: 9653 faults for FIFO and 9651 for LRU, with fault rates remaining 0.967 even after all 10,000 references. OPT performs better at this memory size, finishing with 7773 total faults and reducing its fault rate from 0.826 at 2000 references to 0.777 at 10,000. At 256 frames FIFO and LRU again behave almost identically with 9334 and 9331 total faults, respectively, and both show only slight improvement compared to the 128 frame configuration. Their final fault rates remain high, at about 0.933 for both algorithms. Meanwhile, OPT improves substantially, dropping to 6934 total faults and achieving a much lower final fault rate of 0.693. We see that as memory increases, OPT benefits more dramatically than FIFO or LRU. When increasing the memory further to 512 frames, FIFO and LRUare still almost identical, each producing 8752 total faults, with final fault rates of 0.875. OPT,shows another large improvement as its total faults fall to 5909, and its final fault rate drops to 0.591. Finally, at 1024 frames, all algorithms improve, but OPT continues to outperform the others. FIFO performed 7604 faults with a final fault rate of 0.760, while LRU performed 7585 faults and had a rate of 0.758. OPT drops to 4735 total faults,with a page fault rate of 0.473.

Conclusion

Overall the lab demonstrated the expected outcome from these paging algorithms. OPT has the lowest page fault rates followed by LRU and last being FIFO. From the results we see that OPT has the greatest benefit of increased page frames but since it is not practical in implementation as perfect knowledge of the future is never known it is a good bench mark.

Page Fault Rate vs Reference Count